

Week 08: Tools and controlled actions

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Implement a bounded read-only tool and reason about approval for actions.

Context and teaching plan

The assistant calculates a permitted reimbursement cap; a request to update a record needs approval.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Tool

A named operation with validated arguments, permissions and a defined result.

Teacher question: Where does tool appear in the weekly case, and what could go wrong if we omit it?

Workflow

Code chooses the next step by a fixed rule.

Teacher question: Where does workflow appear in the weekly case, and what could go wrong if we omit it?

Agent

A model may choose a next action, within the permissions and limits supplied by the application.

Teacher question: Where does agent appear in the weekly case, and what could go wrong if we omit it?

Approval boundary

A consequential write needs explicit review before execution and an audit record.

Teacher question: Where does approval boundary appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Policy cap is €40; eligible expense is €55.
- Change or question: Read-only tool returns €40.
- Supported observation: A request to change the policy cap is denied until the authorized reviewer approves.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The reimbursement cap tool: A €55 claim has a €40 cap. The assistant may calculate the payable amount. A separate request asks it to change the cap. Which path needs approval?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Tool: analogy, mechanism and check

Familiar picture: A calculator performs one defined operation when given a valid number.

Familiar part	Technical meaning
Number	tool argument
Guard	input validator
Display	tool result

Where this analogy breaks: The model may choose the wrong tool or arguments; inspect the call.

Actual idea: A named operation with validated arguments, permissions and a defined result.

Work it aloud: `cap(55,40)` returns €40; `cap(-5,40)` is rejected as invalid input.

Ask before revealing: What should run before the arithmetic?

Supported response: Type and range validation.

Look for this misconception: A tool call is safe because its function is simple.

Workflow: analogy, mechanism and check

Familiar picture: An assembly line follows the same stations in a fixed order.

Familiar part	Technical meaning
Station	code step
Conveyor	workflow

Finished item	checked result
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Where this analogy breaks: Real workflows can branch on validated conditions, while remaining predefined.

Actual idea: Code chooses the next step by a fixed rule.

Work it aloud: Validate amount → calculate cap → format reply; a negative amount stops at validation.

Ask before revealing: Who chooses the next station in this version?

Supported response: The predefined workflow code.

Look for this misconception: A workflow must involve a free-form agent.

Agent: analogy, mechanism and check

Familiar picture: A dispatcher selects among allowed services to handle a request.

Familiar part	Technical meaning
Dispatcher	model decision
Service desk	tool interface
Ticket	logged result

Where this analogy breaks: Agent choice is bounded by available tools and permissions; autonomy is limited.

Actual idea: A model may choose a next action, within the permissions and limits supplied by the application.

Work it aloud: The model proposes calculator; the interface validates €55 before executing it.

Ask before revealing: Can the model authorize a write by choosing a tool?

Supported response: No. Authorization is enforced outside the model.

Look for this misconception: An agent can grant itself a new permission.

Approval boundary: analogy, mechanism and check

Familiar picture: A bank transfer waits for an authorized person to approve it.

Familiar part	Technical meaning
Transfer request	write action
Approver	authorized reviewer
Receipt	audit record

Where this analogy breaks: Some read-only actions also need controls when they reveal restricted information.

Actual idea: A consequential write needs explicit review before execution and an audit record.

Work it aloud: Changing the cap requires an approval record; calculation of €40 does not change the record.

Ask before revealing: What evidence shows a write was authorized?

Supported response: An approval tied to the request and recorded action.

Look for this misconception: A polite request is enough approval.

New situation for discussion

A tool receives amount -€5. What should happen before any calculation?

Memory cue: Validate a read-only tool; route write actions through human approval.

Guided laboratory

Implement a calculator and approval gate simulation.

1. Call a read-only capped calculator
2. Reject a negative or non-numeric amount
3. Simulate an approval request without a real write
4. Check that a denied request remains unchanged

Individual change: Try a negative amount and confirm the validator blocks it. Expected evidence: Tool result, blocked-request trace and approval explanation.

Checks and common error

Measure: Unauthorized successful actions / attempted unauthorized actions (target zero). Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: A tool call chosen by a model is automatically authorized.

Correction: The application checks role, arguments and approval independently of the model text.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Anthropic, Building effective agents. <https://www.anthropic.com/engineering/building-effective-agents>

OpenAI, Agent safety. <https://developers.openai.com/api/docs/guides/agent-builder-safety>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Finetune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO4, LO5 | 2 hours concepts + 2 hours guided practice

The idea in plain language

A tool has a name, expected arguments and a bounded operation. A workflow follows fixed code paths. An agent may choose its next step, but permissions and argument validation still belong to the application. Read-only calculations and record changes require different controls.

A worked example

An eligible expense of 55 with a cap of 40 gives $\min(55, 40) = 40$. Negative or nonfinite amounts are invalid. A separate mock request to change a payroll record must remain blocked until a trusted manager approval is recorded.

Practice before reading the answers

1. What distinguishes a workflow from an agent? (4 suggested practice marks)
2. Calculate a reimbursement with eligible expense 55 and cap 40. (4 suggested practice marks)
3. A model says approved. Is this enough to write a record? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. A workflow follows predefined steps; an agent can select subsequent actions within allowed boundaries.
2. $\min(55,40)=40$. Eligibility must already be established; the calculator does not decide it.
3. No. A trusted approval check must authorize the specific action and arguments.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

O2 OpenAI: Safety in building agents

<https://developers.openai.com/api/docs/guides/agent-builder-safety>

Study focus: Tool approvals, untrusted input and controls.

N1 NVIDIA: Generative AI and LLM Learning Paths

<https://www.nvidia.com/en-us/learn/learning-path/generative-ai-llm/>

Study focus: Optional deeper RAG, diffusion and deployment training; some courses require payment.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.